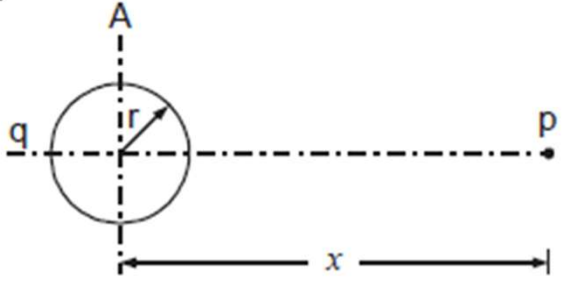
Potential of a Charged Single Conductor

- Consider a long cylindrical conductor of radius r carrying a uniform charge q C/m.

- The potential of the conductor is obtained by integrating the electric field from the conductor surface to infinity.

$V_A = \int_r^\infty E\, dx$

$V_A = (q / 2\pi \varepsilon_0) \int_r^\infty (dx/x)$



$$V_A = \int_r^\infty \frac{q}{2\pi \epsilon_0 x} dx = \frac{q}{2\pi \epsilon_0} \int_r^\infty \frac{dx}{x}$$